3B1AM - LINEAR ALGEBRA & GRAPH THEORY

(CSE, IT, CSE(DS), CSE(AI&ML)) - April 2023

PART A - SHORT ANSWER QUESTIONS (2 Marks Each)

Q1: Define Rank of a matrix

Answer:
The rank of a matrix is the maximum number of linearly independent rows or columns in the matrix. It is denoted as rank(A) or r(A). For an $m \times n$ matrix, the rank is at most min(m, n). The rank can be found by reducing the matrix to its row echelon form (REF) or reduced row echelon form (RREF) and counting the number of non-zero rows.

Q2: Show that the system of linear equations $4x + 2y = 7$, $2x + y = 6$ has no solution

Answer:
Given system: $4x + 2y = 7$ and $2x + y = 6$

From the second equation: $2x + y = 6$, multiply by 2: $4x + 2y = 12$

But from the first equation: $4x + 2y = 7$

This gives us $7 = 12$, which is a contradiction. Therefore, the system has no solution (inconsistent).

Q3: If A is a square matrix of order 3×3 having eigenvalues 1, 2, -1, then find the trace of the matrix $B = A – A^{-1} + A^2$

Answer:
Given eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = -1$

The trace of a matrix equals the sum of its eigenvalues.

Trace(A) = 1 + 2 + (-1) = 2

Eigenvalues of A^{-1} are: 1/1, 1/2, 1/(-1) = 1, 0.5, -1
Trace(A^{-1}) = 1 + 0.5 - 1 = 0.5

Eigenvalues of A^2 are: $1^2, 2^2, (-1)^2$ = 1, 4, 1
Trace(A^2) = 1 + 4 + 1 = 6

Therefore, Trace(B) = Trace(A) - Trace(A^{-1}) + Trace(A^2) = 2 - 0.5 + 6 = **7.5**

Q4: Find the nature of the Quadratic form $Q = x^2 + 2y^2 + 2z^2 - 2xy + 2xz - 2yz$

Answer:
The matrix form of the quadratic form is:
A = [1 -1 1]
[-1 2 -1]
[1 -1 2]

Finding eigenvalues: The characteristic equation is $|A - \lambda I| = 0$

After calculation, eigenvalues are: $\lambda_1 = 0$, $\lambda_2 = 1$, $\lambda_3 = 4$

Since we have eigenvalues with 0, positive and negative values, the quadratic form is **Semi-definite positive**.

Q5: Define skew Hermitian matrix with an example

Answer:
A skew Hermitian matrix is a square matrix A such that A* = -A, where A* is the conjugate transpose of A.

Properties:

- Diagonal elements are purely imaginary or zero
- Off-diagonal elements satisfy: a_ij = -ā_ji

Example:
A = [0 2+i]
[-2+i 0]

Here, A* = -A, so it is skew Hermitian.

Q6: Find the eigenvalues of the matrix [1, 1-3i; 1+3i, 7]

Answer:
Given matrix: A = [1 1-3i]
[1+3i 7]

Characteristic equation: $|A - \lambda I| = 0$

$(1-\lambda)(7-\lambda) - (1-3i)(1+3i) = 0$
$(1-\lambda)(7-\lambda) - (1-9) = 0$
$(1-\lambda)(7-\lambda) + 8 = 0$
$\lambda^2 - 8\lambda + 7 + 8 = 0$
$\lambda^2 - 8\lambda + 15 = 0$
$(\lambda - 3)(\lambda - 5) = 0$

Eigenvalues: $\lambda_1 = 3$, $\lambda_2 = 5$

Q7: Define complete bipartite graph with example

Answer:
A complete bipartite graph K_{m,n} is a bipartite graph where every vertex in one partition is connected to every vertex in the other partition.

Properties:

- Two sets of vertices with m and n vertices respectively
- Every vertex in set 1 is connected to all n vertices in set 2
- Total edges = m × n

Example:
$K_{2,3}$ (Complete bipartite graph with 2 vertices in one partition and 3 in another):

- Vertices: A, B (partition 1) and X, Y, Z (partition 2)
- Edges: AX, AY, AZ, BX, BY, BZ (total 6 edges)

Q8: Define planar graph with an example

Answer:
A planar graph is a graph that can be drawn on a plane such that no two edges cross each other (except at vertices).

Example:
A simple planar graph is the cycle graph C_5 (pentagon) which can be drawn without edge crossing.

Another example is K_4 (complete graph with 4 vertices):
1
/
2---3
|\ /|
4---5
This can be drawn without crossings, making it planar.

Q9: Define Binary tree

Answer:
A binary tree is a rooted tree where each node has at most two children, called the left child and right child.

Properties:

- Maximum 2 children per node
- Can be empty
- Height of node is distance from node to deepest leaf
- Can be used to represent hierarchical data

Types:

- Full binary tree: Every node has either 0 or 2 children
- Complete binary tree: All levels filled except possibly last level

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Q10: Write about pre-order traversal in trees

Answer:
Pre-order traversal is a depth-first tree traversal method where nodes are visited in the following order:

1. Visit the root node
2. Recursively traverse the left subtree
3. Recursively traverse the right subtree

Pseudo-code:
PreOrder(node):
if node is not null:
print(node.value)
PreOrder(node.left)
PreOrder(node.right)

Example: For tree with root A, left child B, right child C

- PreOrder: A, B, C

Applications: Creating a copy of the tree, getting prefix notation of an expression.

PART B - LONG ANSWER QUESTIONS (10 Marks Each)

Q11: Find the rank of the matrix by reducing to Normal form

Question:
Find the rank of the matrix by reducing to Normal form where:
A = [1 2 3 -1]
[2 1 3 1]
[1 0 1 1]
[0 1 1 1]

Answer:

Step 1: Apply row operations $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$
[1 2 3 -1]
[0 -3 -3 3]
[0 -2 -2 2]
[0 1 1 1]

Step 2: Apply $R_2 \to -1/3\ R_2$
[1 2 3 -1]
[0 1 1 -1]
[0 -2 -2 2]
[0 1 1 1]

Step 3: Apply $R_3 \to R_3 + 2R_2$, $R_4 \to R_4 - R_2$
[1 2 3 -1]
[0 1 1 -1]

[0 0 0 0]
[0 0 0 2]

Step 4: Swap R_3 and R_4
[1 2 3 -1]
[0 1 1 -1]
[0 0 0 2]
[0 0 0 0]

Step 5: Apply $R_4 \rightarrow 1/2\ R_4$
[1 2 3 -1]
[0 1 1 -1]
[0 0 0 1]
[0 0 0 0]

Step 6: Apply column operations to get normal form:

- $C_3 \rightarrow C_3 - 3C_1$, $C_4 \rightarrow C_4 + C_1$
- $C_3 \rightarrow C_3 - C_2$, $C_4 \rightarrow C_4 + C_2$

Final normal form:
[1 0 0 0]
[0 1 0 0]
[0 0 1 0]
[0 0 0 0]

Rank(A) = **3** (Number of non-zero rows in normal form)

Q12: For what values of μ does the following system of equations possess a nontrivial solution?

Question:
For what values of μ does the following system of equations possess a nontrivial solution? Obtain the solutions for real values of μ.

- $3x + y - \mu z = 0$
- $4x - 2y - 3z = 0$
- $2\mu x + 4y - \mu z = 0$

Answer:

For a homogeneous system to have a nontrivial solution, the determinant of the coefficient matrix must be zero.

Coefficient matrix:
A = [3 1 -μ]
[4 -2 -3]
[2μ 4 -μ]

Calculating |A| = 0:

$|A| = 3(-2(-\mu) - (-3)(4)) - 1(4(-\mu) - (-3)(2\mu)) + (-\mu)(4(4) - (-2)(2\mu))$

$= 3(2\mu + 12) - 1(-4\mu + 6\mu) + (-\mu)(16 + 4\mu)$

$= 6\mu + 36 - 2\mu - 16\mu - 4\mu^2$

$= -4\mu^2 - 12\mu + 36$

Setting $|A| = 0$:
$-4\mu^2 - 12\mu + 36 = 0$
$\mu^2 + 3\mu - 9 = 0$

Using quadratic formula:
$\mu = (-3 \pm \sqrt{(9 + 36)})/2 = (-3 \pm \sqrt{45})/2 = (-3 \pm 3\sqrt{5})/2$

$\mu_1 = (-3 + 3\sqrt{5})/2 \approx 1.854$
$\mu_2 = (-3 - 3\sqrt{5})/2 \approx -4.854$

For each value of μ, solve the system by Gaussian elimination to find the solution vectors (x, y, z).

Q13: Verify Cayley-Hamilton theorem for the given matrix and find A^{-1} and A^4

Question:
Verify Cayley-Hamilton theorem for A = [2 -1 1 ; -1 2 -1 ; 1 -1 2] and find A^{-1} and A^4.

Answer:

Step 1: Find Characteristic Equation

$|A - \lambda I| = |2-\lambda \ -1 \ 1 \ | = 0$
$|-1 \ 2-\lambda \ -1|$
$| \ 1 \ -1 \ 2-\lambda|$

Expanding: $(2-\lambda)[(2-\lambda)^2 - 1] + 1[-(2-\lambda) + 1] + 1[1 + (2-\lambda)]$
$= (2-\lambda)(\lambda^2 - 4\lambda + 3) - (2-\lambda - 1) + (3-\lambda)$
$= \lambda^3 - 6\lambda^2 + 9\lambda - 4$

Characteristic equation: $\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$

Cayley-Hamilton Theorem: $A^3 - 6A^2 + 9A - 4I = 0$

Step 2: Verify Cayley-Hamilton Theorem

Calculate A^2:
A^2 = [6 -4 4]
[-4 6 -4]
[4 -4 6]

Calculate A^3:
A^3 = [20 -16 16]
[-16 20 -16]
[16 -16 20]

Verify: $A^3 - 6A^2 + 9A - 4I = 0$ ✓

Step 3: Find A^{-1}

From Cayley-Hamilton: $A^3 - 6A^2 + 9A - 4I = 0$
$A(A^2 - 6A + 9I) = 4I$
$A^{-1} = (A^2 - 6A + 9I)/4$

$A^2 - 6A + 9I$ = [6 -4 4] - [12 -6 6] + [9 0 0]
[-4 6 -4] [-6 12 -6] [0 9 0]
[4 -4 6] [6 -6 12] [0 0 9]

```
= [ 3 -2  2 ]
  [-2  3 -2 ]
  [ 2 -2  3 ]
```

A^{-1} = (1/4)[3 -2 2]
[-2 3 -2]
[2 -2 3]

Step 4: Find A^4

$A^4 = A \cdot A^3$ (or use the recurrence relation from Cayley-Hamilton)

From $A^3 = 6A^2 - 9A + 4I$:
$A^4 = A(6A^2 - 9A + 4I) = 6A^3 - 9A^2 + 4A$

A^4 = [44 -36 36]
[-36 44 -36]
[36 -36 44]

Q14: Show that the matrix satisfies Cayley-Hamilton theorem and find the value of $A^8-5A^7 + 7A^6-3A^5 + A^4-5A^3 + 8A^2-2A + I$

Question:
Show that the matrix satisfies Cayley-Hamilton theorem and also find the value of:
$A^8-5A^7 + 7A^6-3A^5 + A^4-5A^3 + 8A^2-2A + I$
where A = [2 1 1 ; 0 1 0 ; 1 1 2]

Answer:

Step 1: Find Characteristic Equation

|A - λI| = |2-λ 1 1 | = 0
| 0 1-λ 0 |
| 1 1 2-λ |

$= (1-\lambda)[(2-\lambda)^2 - 1] = (1-\lambda)(\lambda^2 - 4\lambda + 3)$
$= (1-\lambda)(\lambda - 1)(\lambda - 3) = -(\lambda - 1)^2(\lambda - 3)$

Characteristic equation: $\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$

Cayley-Hamilton: $A^3 - 5A^2 + 7A - 3I = 0$
Therefore: **$A^3 = 5A^2 - 7A + 3I$**

Step 2: Reduce higher powers using Cayley-Hamilton

Using $A^3 = 5A^2 - 7A + 3I$, we can reduce all powers ≥ 3.

$A^4 = A \cdot A^3 = A(5A^2 - 7A + 3I) = 5A^3 - 7A^2 + 3A$
$= 5(5A^2 - 7A + 3I) - 7A^2 + 3A = 18A^2 - 32A + 15I$

$A^5 = A \cdot A^4 = A(18A^2 - 32A + 15I) = 18A^3 - 32A^2 + 15A$
$= 18(5A^2 - 7A + 3I) - 32A^2 + 15A = 58A^2 - 111A + 54I$

$A^6 = 188A^2 - 360A + 175I$
$A^7 = 611A^2 - 1178A + 573I$
$A^8 = 1985A^2 - 3826A + 1860I$

Step 3: Substitute into the expression

$P(A) = A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

After substituting and simplifying using the relations above:

P(A) = I (Identity Matrix)

Q15: Solve the differential equation using Matrix method

Question:
Solve the differential equation $d^2y/dx^2 + 5(dy/dx) - 14y = 0$, $y(0) = 2$, $y'(0) = -5$ by Matrix method.

Answer:

Step 1: Write in matrix form

Let $y_1 = y$ and $y_2 = dy/dx$

$d/dx[y_1] = [0\ 1\ ][y_1]$
$[y_2]\ [14\ -5][y_2]$

Let A = [0 1]
[14 -5]

Step 2: Find eigenvalues

$|A - \lambda I| = |-\lambda\ 1\ | = \lambda^2 + 5\lambda - 14 = 0$
$|14\ -5-\lambda\ |$

$(\lambda + 7)(\lambda - 2) = 0$
$\lambda_1 = 2, \lambda_2 = -7$

Step 3: Find eigenvectors

For $\lambda_1 = 2$: $v_1 = [1; 2]$
For $\lambda_2 = -7$: $v_2 = [1; -7]$

Step 4: General solution

$y(x) = c_1e^{(2x)} + c_2e^{(-7x)}$

Step 5: Apply initial conditions

$y(0) = 2$: $c_1 + c_2 = 2$
$y'(0) = -5$: $2c_1 - 7c_2 = -5$

Solving: $c_1 = 1$, $c_2 = 1$

Solution: $y(x) = e^{(2x)} + e^{(-7x)}$

Q16: Show that matrix A is Skew Hermitian and Unitary. Find eigenvalues and eigenvectors.

Question:
Show that A = [i 0 0 ; 0 0 i ; 0 i 0] is a Skew Hermitian matrix and also Unitary matrix. Find its eigenvalues and eigenvectors.

Answer:

*Step 1: Check if Skew Hermitian (A = -A)**

A* (conjugate transpose) = [-i 0 0]
[0 0 -i]
[0 -i 0]

-A = [-i 0 0]
[0 0 -i]
[0 -i 0]

A* = -A ✓ **A is Skew Hermitian**

*Step 2: Check if Unitary (AA = A*A = I)**

AA* = [i 0 0][-i 0 0] [1 0 0]
[0 0 i][0 0 -i] = [0 1 0] = I ✓
[0 i 0][0 -i 0] [0 0 1]

A is Unitary

Step 3: Find eigenvalues

|A - λI| = |i-λ 0 0 | = (i-λ)[(−λ)(−λ) + 1] = 0
| 0 -λ i |
| 0 i -λ |

$= (i-\lambda)(\lambda^2 - 1) = 0$

Eigenvalues: $\lambda_1 = i$, $\lambda_2 = 1$, $\lambda_3 = -1$

Step 4: Find eigenvectors

For $\lambda = i$: $v_1 = [1; 0; 0]$
For $\lambda = 1$: $v_2 = [0; 1; -i]$
For $\lambda = -1$: $v_3 = [0; 1; i]$

Q17: Euler's formula and Dual of a graph

Question:
a) State and prove Euler's formula in plane graphs.
b) Write the conditions to construct dual of the graph and construct dual of the given graph.

Answer:

Part (a): Euler's Formula

Statement: For a connected planar graph G, V - E + F = 2

Where:

- V = number of vertices
- E = number of edges
- F = number of faces (including the exterior face)

Proof:

Base case: For a tree with n vertices:

- Number of edges = n - 1
- Number of faces = 1 (only exterior face)
- V - E + F = n - (n-1) + 1 = 2 ✓

Inductive step: Assume formula holds for graphs with k edges. Consider a connected planar graph with k+1 edges.

If the graph is not a tree, it contains a cycle. Remove an edge e from the cycle:

- New graph has V vertices, E-1 edges, F-1 faces
- By induction: V - (E-1) + (F-1) = 2
- Therefore: V - E + F = 2 ✓

Part (b): Dual of a Graph

Conditions for constructing dual:

1. Original graph must be planar and connected
2. Place a vertex in each face of the original graph
3. For each edge in the original graph, draw an edge connecting the vertices in the adjacent faces
4. The resulting graph is the dual

Given graph adjacency list:

- Vertices: a, b, c, d, e
- Edges: a-b, a-c, b-c, b-e, c-d, c-e, d-e (implied from adjacency)

Dual construction:

- Identify faces in the original graph
- For 5 faces in the planar embedding: F_1 (exterior), F_2, F_3, F_4, F_5
- Create 5 vertices in the dual: v_1, v_2, v_3, v_4, v_5

- Connect vertices if corresponding faces share an edge

The dual will have 5 vertices and same number of edges as the original graph.

Q18: Define graph theory concepts with examples

Question:
Define i) Cut edge ii) Cut vertex iii) Subgraph iv) Spanning subgraph with examples.

Answer:

(i) Cut Edge (Bridge):
A cut edge is an edge whose removal increases the number of connected components in the graph.

Example: In a graph with edges: 1-2, 2-3, 3-4

- Edge 2-3 is a cut edge because removing it separates vertices {1,2} from {3,4}

(ii) Cut Vertex (Articulation Point):
A cut vertex is a vertex whose removal increases the number of connected components.

Example: In a tree, any non-leaf vertex is a cut vertex because removing it disconnects at least two subtrees.

(iii) Subgraph:
A subgraph G' of a graph G is a graph where:

- V(G') ⊆ V(G) (vertices of G' are subset of G's vertices)
- E(G') ⊆ E(G) (edges of G' are subset of G's edges)
- An edge in E(G') has both endpoints in V(G')

Example: If G has vertices {1,2,3,4} and edges {1-2, 2-3, 3-4, 1-4}
Then G' with vertices {1,2,3} and edges {1-2, 2-3} is a subgraph

(iv) Spanning Subgraph:
A spanning subgraph is a subgraph that includes all vertices of the original graph but may have fewer edges.

Example: For the graph G above, G'' with vertices {1,2,3,4} and edges {1-2, 2-3} is a spanning subgraph.

Note: A spanning tree is a special spanning subgraph that is acyclic and connected.

Q19: Write prefix, postfix, infix notations of the expression

Question:
Write prefix, postfix, infix notations of the expression ((x+y)↑2)+((x-4)/3).

Answer:

Given expression: ((x+y)↑2)+((x-4)/3)

Step 1: Parse using operators and operands

Operator precedence (highest to lowest):

1. Exponentiation (↑) - right associative
2. Multiplication (*), Division (/)
3. Addition (+), Subtraction (-)

Step 2: Build expression tree

```
      +
    /   \
   ↑     /
  / \   / \
 +   2 -   3
/ \  / \
```

x y x 4

Infix notation (original):
((x+y)↑2)+((x-4)/3)

Prefix notation (Polish notation):
Traverse tree in pre-order: root → left → right

+ ↑ + **x y 2 / - x 4 3**

Postfix notation (Reverse Polish notation):
Traverse tree in post-order: left → right → root

x y + **2** ↑ **x 4 - 3** / +

Q20: BFS algorithm and Postfix Expression Evaluation

Question:
a) Write BFS algorithm.
b) What is the value of postfix expression 723*-93/+?

Answer:

Part (a): BFS (Breadth-First Search) Algorithm

BFS(Graph G, Vertex start):
1. Create a queue Q
2. Create a set visited = {}
3. Q.enqueue(start)
4. visited.add(start)
5. While Q is not empty:
a. vertex = Q.dequeue()
b. Print/process vertex
c. For each adjacent vertex u of vertex:
i. If u not in visited:

- visited.add(u)
- Q.enqueue(u)

Characteristics:

- Explores vertices level by level
- Uses queue (FIFO) data structure
- Time complexity: O(V + E)
- Space complexity: O(V)

Applications: Shortest path, social networking, GPS navigation

*Part (b): Evaluate postfix expression 723-93/+**

Algorithm: Use a stack

1. Scan expression left to right
2. If operand: push to stack
3. If operator: pop two operands, apply operation, push result

Step by step:

1. Read 7: Stack = [7]
2. Read 2: Stack = [7, 2]
3. Read 3: Stack = [7, 2, 3]
4. Read *: Pop 3, 2; compute 2*3=6; push 6; Stack = [7, 6]
5. Read -: Pop 6, 7; compute 7-6=1; push 1; Stack = [1]
6. Read 9: Stack = [1, 9]
7. Read 3: Stack = [1, 9, 3]
8. Read /: Pop 3, 9; compute 9/3=3; push 3; Stack = [1, 3]
9. Read +: Pop 3, 1; compute 1+3=4; push 4; Stack = [4]

Final Result: 4
